

KnoDL as a Tool for Harmonizing Reference and Regulatory Data in Cross-Border Transport Systems

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Abstract: [Objective] The purpose of this article is to show the direction of eliminating significant barriers in digital international trade and logistics caused by inconsistencies in reference and regulatory data by proposing KnoDL, a declarative mapping technology designed to create a single digital layer for cross-border data exchange [1, 2]. The proposed solution uses a formal, rules-based approach for harmonization and standardization heterogeneous reference systems and classifiers, without relying on machine learning models. This method provides deterministic, reproducible, and legally sound data transformations that can be integrated into existing IT infrastructures without significant changes. The implementation of KnoDL demonstrates tangible benefits, including a 30-70% reduction in document preparation time, elimination of data conflicts and duplicate records, as well as increased automation. A comparative analysis shows that this formal approach surpasses ML/AI solutions in terms of explainability, legal verifiability, and resource efficiency [3].

KnoDL serves as a key technological element for creating a reliable digital ecosystem for multimodal transportation, ensuring compatibility with international standards such as eFTI, eCMR and GS1. Its transparent and legally sound methodology makes it the preferred choice for regulated logistics processes where accuracy and provability of data are crucial [4].

Keywords: data harmonization; digital logistics; formal methods; KnoDL; cross-border transportation; data fragmentation; UN/CEFACT

1. Introduction: The Invisible Barrier in Digital Logistics

The digital transformation of global logistics is hindered by a critical, yet often underestimated, problem: data fragmentation. A typical multimodal freight journey involves numerous participants—road carriers, railways, ports, and customs authorities—each operating with their own siloed data systems, document standards, and cargo classifiers. This disjointed landscape forces manual data "translation" at each stage, introducing costly delays, the risk of critical errors, and preventing true automation. International bodies like UNECE/UN/CEFACT recognize this challenge, noting that efforts to create unified semantic models (e.g., MMT RDM) are hamstrung without tools capable of seamlessly bridging these data silos [5, 6]. Data fragmentation is not a mere technicality but a strategic bottleneck undermining the speed, reliability, and economics of the entire transport system.

2. Advantages of Formal Methods over ML/AI Approaches

To address the need for accuracy and transparency in mission-critical logistics systems, formal, rule-based methods offer distinct advantages over machine learning (ML/AI) approaches. While ML models are efficient at processing unstructured data, their application is limited by factors such as operating as a "black box," difficulty in interpreting results, and the need for large training datasets and computational resources. Formal methods, in contrast, provide full explainability and reproducibility, require low resources, and are easily standardized. This makes them highly effective in scenarios

where legally accurate and reproducible data matching is paramount, such as in digital logistics where regulatory fragmentation leads to tangible costs.

3. The KnoDL Solution

KnoDL (Knowledge Definition Language) [7] is a lightweight module designed to solve data standardization challenges by embodying the principles of applicable formal methods. It is a formal data matching and unification system based on rules rather than model training. Its key functions include:

- **Automated Matching:** Automatically aligning entries across different reference databases and product catalogs, even with varying structures and formats.
- **Data Standardization:** Standardizing regulatory and reference data into a unified format without modifying existing IT systems.
- **Compliance Verification:** Proactively comparing cargo characteristics against current regulations across multiple transport modes to identify potential non-compliance before shipment.
- **Traceability:** Allowing for full traceability of every data transformation, ensuring complete transparency.

4. Results and Impact

A significant number of experiments by the authors and pilot applications conducted by users using KnoDL modules on various datasets show that the implementation of KnoDL provides scalable and measurable results not only for cross-border logistics, but also for other related tasks such as standard alignment, technology search and evaluation, ontology development, analysis and interpretation of big data for dynamic ITS models, etc. [8, 9]. Automated comparison of reference data and regulatory documents reduces the time required for the preparation of transport and regulatory documents by 30-70% [10].

Tables 1-3 show experimental data on processing speed for small, medium, and large-volume data arrays.

Table 1 Processing speed for Small Datasets (1k-50k records)

Use Case	Data Volume	Processing Time	Hardware	Throughput
Tourism vs IT References	38k rows, 8.5 MB	35 seconds	Intel i7-1165G7, 16GB	2,5M rows/sec
Wine Classification	1,599 records	<1 second	Razer 7, 16GB RAM	60K rows/sec
Twitter Sentiment	14,640 records	1 min 50 sec	Razer 7, 16GB RAM	2M rows/sec
LMS Accounting	25k vs 30k rows, 40 MB	3 min 05 sec	AWS m5.4xlarge	4M rows/sec

Table 2 Processing speed for Small Datasets (50k-1M records)

Use Case	Data Volume	Processing Time	Hardware	Throughput
Product Catalogs	40k vs 300k rows, 95.4 MB	20 min 42 sec	Intel i7-1165G7, 16GB	9,5M rows/sec
1C ERP Integration	120k vs 110k rows, 97 MB	6 min 55 sec	AWS m5.4xlarge, 64GB	31M rows/sec
Warehouse Consolidation	900k rows, 240 MB	11 min 30 sec	AWS m5.4xlarge, 64GB	1,3B rows/sec
Corporate	200k rows, 35	2 min 10 sec	MacBook Pro	300M rows/sec

Table 3 Processing speed for Large Datasets (1M-30M records)

Use Case	Data Volume	Processing Time	Hardware	Throughput
MES Melt Log Processing	3M rows, 850 MB	46 min 18 sec	AWS c5n.4xlarge, 42GB	3B rows/sec
Digital Production Model	4M rows, 1.1 GB	52 min 40 sec	AWS c5n.4xlarge, 42GB	5B rows/sec
Unstructured Reports	5M rows, 1 GB	3h 05 min	RHEL, Xeon Silver 12-core	2B rows/sec
Regulatory Records	30M rows, 6 GB	17h 04 min	Linux VDS, 8 vCPU, 32GB	14B rows/sec

Processes that previously took weeks of manual reconciliation can be completed in hours. The system effectively eliminates duplicate records and logical conflicts that arise from combining national and industry-specific data, thereby increasing the reliability and legal verifiability of documents. This formal, deterministic approach builds a key element of a trustworthy digital ecosystem for multimodal transport, aligning with the goals of international organizations to create a seamless and efficient global logistics network.

Conclusion:

In conclusion, it should be noted that with the DSL (Domain Specific Language) module, a domain—specific language that has become an effective and simple constructor for describing the logic of various processes, the KnoDL system is turning into a platform for analysts and system architects that supports the digitization of technological and business processes.

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